Lecture - 40

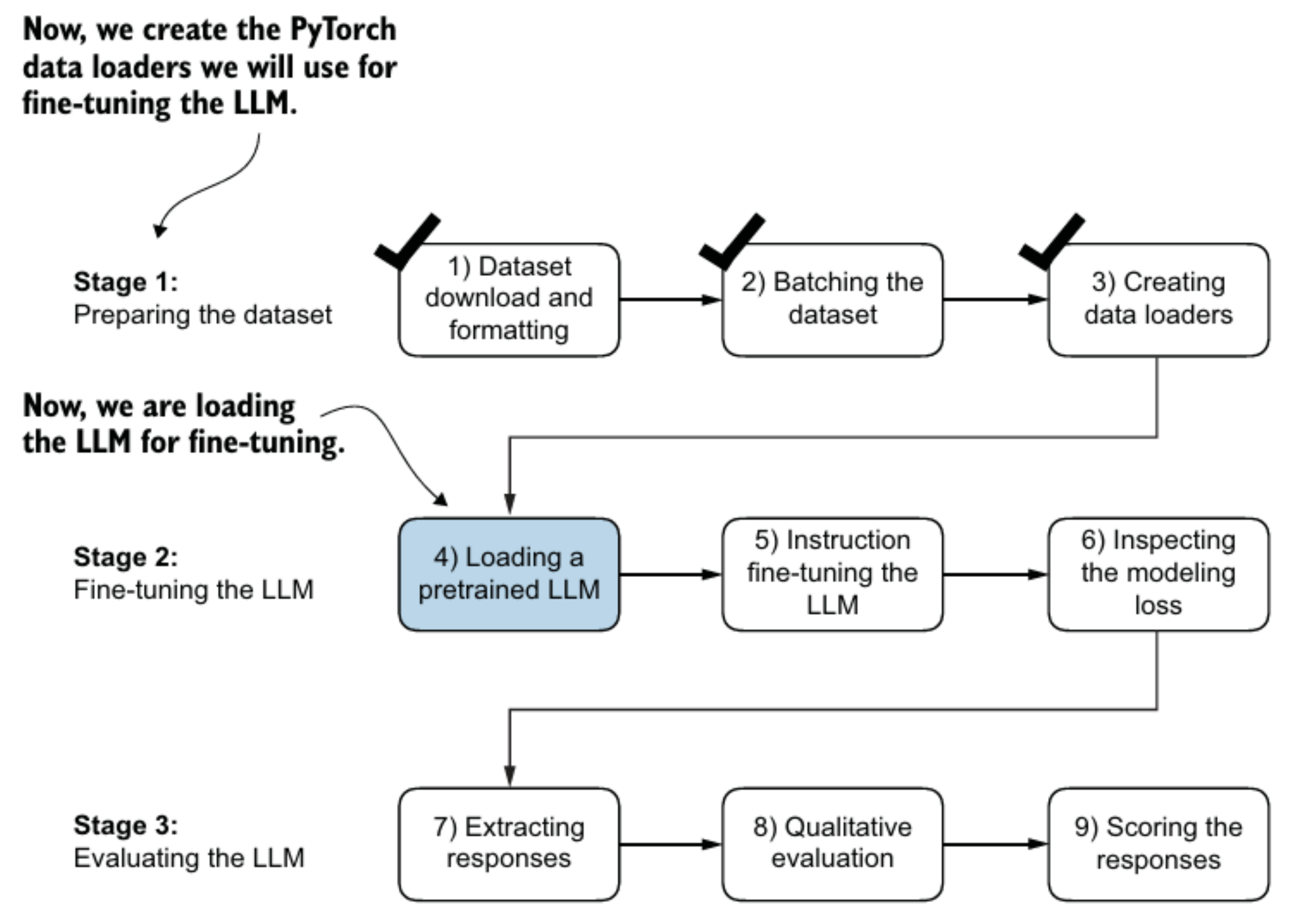


Figure 7.15 The three-stage process for instruction fine-tuning an LLM. After the dataset preparation, the process of fine-tuning an LLM for instruction-following begins with loading a pretrained LLM, which serves as the foundation for subsequent training.

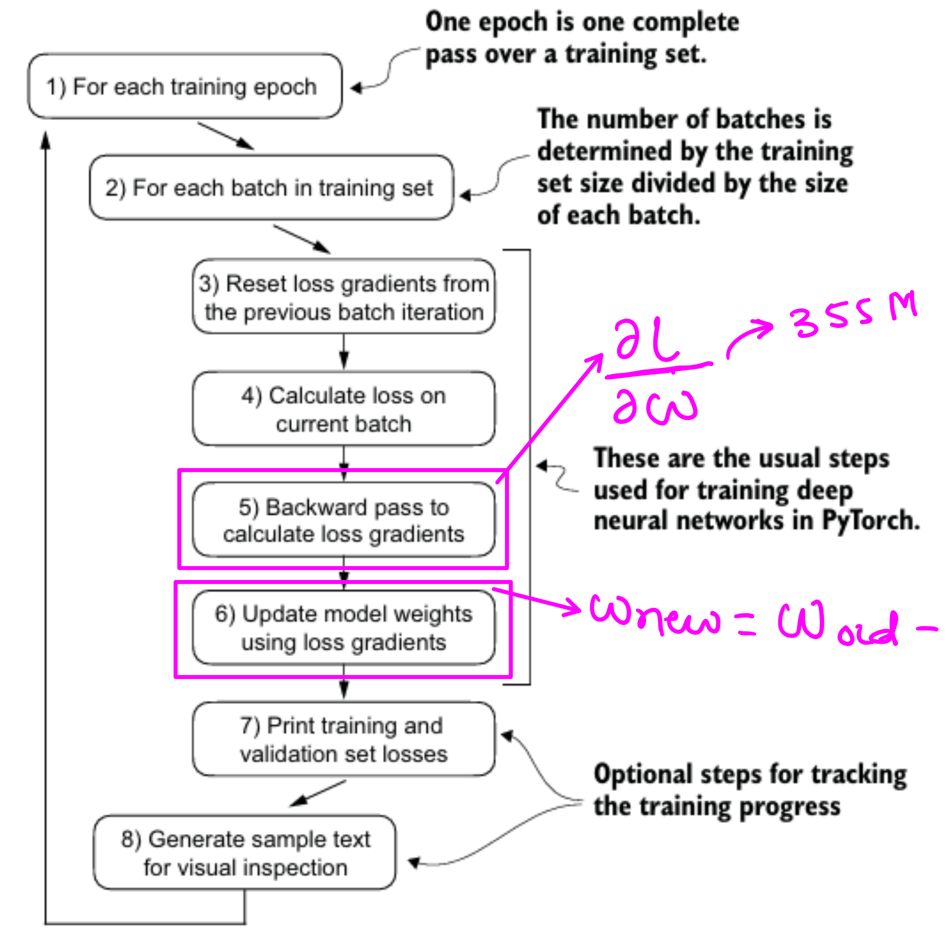
9. Finetuning the model (step-5)

(a) Loss function: cross entropy loss

(similar to the one used for pre-training the LLM)

(b) Run training loop on instruction dataset

- Due to pre-training model starts from knowledge state, instead of random.



$\frac{\partial L}{\partial w} \rightarrow 355M$

$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w}$

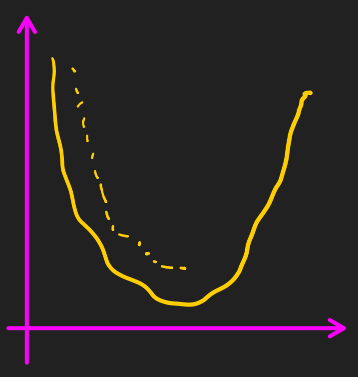


Figure 6.15 A typical training loop for training deep neural networks in PyTorch consists of several steps, iterating over the batches in the training set for several epochs. In each loop, we calculate the loss for each training set batch to determine loss gradients, which we use to update the model weights to minimize the training set loss.